Since you already work with C#, ASP.NET Core, SQL Server, RAG, semantic search, ML.NET, and Gemma, I recommend building one reusable Corporate AI Platform and then developing individual AI applications on top of it.

The goal is to create something your company can demonstrate to management, integrate into existing client systems, and eventually offer as a commercial product.

Corporate AI Development Roadmap

1. The overall development plan

1. Phase 1 — AI fundamentals and architecture

   Week 1–2
   - Understand LLMs, embeddings, RAG, tool calling, and structured output.
   - Decide when to use an LLM versus traditional machine learning.
   - Define privacy, permissions, and evaluation requirements.
2. Phase 2 — Build shared AI services

   Week 3–4
   - Create an ASP.NET Core Web API.
   - Build reusable model integration, prompt handling, and embedding services.
   - Add logging, configuration, error handling, and authentication.
3. Phase 3 — Develop the first AI application

   Week 5–6
   - Build an Enterprise AI Knowledge Assistant.
   - Upload sample documents and implement RAG.
   - Return answers with source references.
   - Test accuracy and document access permissions.
4. Phase 4 — Develop additional AI modules

   Week 7–10
   - Intelligent Document Processing.
   - Customer Support Assistant.
   - Business Intelligence Assistant.
   - Reuse shared components wherever possible.
5. Phase 5 — Advanced AI applications

   Week 11–14** - AI Code Review. - Contract and Compliance Analysis. - Recruitment Matching. - Predictive Analytics and Project Risk Detection.
6. Phase 6 — Pilot and production readiness

   Week 15–16 and ongoing

- Deploy to a controlled test environment.
- Measure quality, speed, cost, and user satisfaction.
- Review security and failure cases.
- Collect feedback and prepare a client demonstration.

These timelines are an illustrative plan for a focused prototype effort, not a guarantee. Building all the applications to production quality would require more time, especially for data integration, security, and validation.

2. Project-by-project implementation roadmap

Here is the order I recommend, based on your existing C# and AI learning.

Order	Project	Main skills to develop
1	Enterprise AI Knowledge Assistant	Embeddings, RAG, document retrieval
2	Intelligent Document Processing	OCR, extraction, validation
3	AI Customer Support Assistant	NLP, classification, RAG
4	AI Business Intelligence Assistant	Text-to-SQL, database security
5	AI Workflow Copilot	Tool calling, APIs, approval workflows
6	AI Recruitment and Skills Matching	Resume parsing, semantic search
7	AI Contract Analyzer	Clause extraction, comparison, citations
8	AI Code Review Assistant	Static analysis, code context, LLM revie
9	AI Project Risk Predictor	Historical data, ML.NET, forecasting
10	AI Supply Chain Optimizer	Demand forecasting, anomaly detection
11	AI Cybersecurity Assistant	Log analysis, anomaly detection
12	AI Visual Inspection	Computer vision, image classification

The first four are particularly useful for building reusable enterprise software capabilities. Projects 9–12 depend more heavily on having suitable historical data, specialized datasets, or access to relevant infrastructure.

3. Recommended technical architecture

Build the common platform first, so each new application doesn't need its own independent AI infrastructure.

Web Applications

MVC / Blazor / client applications

ASP.NET Core Web API

Authentication · Authorization · Application endpoints

LLM Service

Gemma / hosted models

RAG Service

Embeddings / retrieval

Document AI

OCR / extraction

ML Services

Classification / prediction

Data and Infrastructure

SQL Server · Document storage · Vector index · Audit logs · Monitoring

Suggested solution structure

```
CorporateAI.sln
│
├── CorporateAI.Api
├── CorporateAI.Application
├── CorporateAI.Domain
├── CorporateAI.Infrastructure
├── CorporateAI.AI
│   ├── LLM
│   ├── Embeddings
│   ├── RAG
│   ├── DocumentProcessing
│   └── MachineLearning
│
└── CorporateAI.Tests
```

- `Api` : HTTP endpoints and authentication integration.
- `Application` : business workflows and orchestration.
- `Domain` : business entities and rules.

- `Infrastructure`: database, files, and external integrations.
- `AI`: reusable model inference, retrieval, and prediction services.
- `Tests`: automated testing and AI quality evaluation.

Start with a modular monolith rather than microservices. You can split components into separate services later if usage, deployment, or scaling requirements justify it.

4. What to build in your first six weeks

Here is a concrete checklist for the first project.

Knowledge Assistant MVP

0 / 18 tasks

Week 1

Requirements and design

Choose one department and a real business problem

Prepare approved sample documents

Define success criteria and access rules

Week 2

Document processing

Extract text from PDF files

Split text into meaningful chunks

Store document metadata and access permissions

Week 3

Semantic retrieval

Generate embeddings

Store and search document vectors

Evaluate retrieval quality with sample questions

Week 4

Grounded AI answers

Connect retrieved chunks to Gemma or a hosted LLM

Generate answers using retrieved context

Display source references and handle unknown answers

Week 5

API and security

Expose an ASP.NET Core endpoint

Implement authentication and authorization

Add error handling, logging, and audit records

Week 6

Demonstration and evaluation

Build a simple chat interface

Test answer accuracy and access restrictions

Present measured results and limitations to management

Copy roadmap checklist

5. How to measure success

Before calling any AI application complete, define measurable acceptance criteria.

Metric	What to evaluate
Accuracy	Does the application produce correct results on a test set?
Grounding	Can answers be traced to approved source information?
Security	Can users access only the information they are authorized to see?
Performance	What are the response time and throughput under expected load?
Cost	What are the model, infrastructure, and maintenance costs?
Business impact	How much time or manual effort does the solution actually save?

Don't measure success by whether the AI produces fluent answers. Measure whether it reliably solves the intended business problem.